

Course File

Faculty	Dr Prabhu A
Department	CS & IT
School	Applied Computing
Program	BCA – AI
Program Level	6th Semester
Course Title	Natural Language Processing
Course Code	24BCAAI504
Academic Year	2026-27
Semester	6th
Semester Start Date	20.07.2026
Course Coordinator	Dr Prabhu A
Course Coordinator email ID	Prabhu.a@jainuniversity.ac.in
Faculty member Name	Dr Prabhu A
Faculty member designation	Associate Prof.
Faculty member email ID	Prabhu.a@jainuniversity.ac.in

Teaching Learning and Evaluation Plan (TLEP)

I) Program Details

Department	:	Computer Science and Information Technology
School	:	Applied Computing
Program	:	BCA

1. Program Educational Objectives:

PEO 1	Technical Proficiency and Problem-Solving Demonstrate expertise in solving complex problems using modern computing tools
PEO 2	Lifelong Learning and Adaptability Engage in lifelong learning to adapt to technological advancements.
PEO 3	Ethical and Responsible Leadership Exhibit ethical behaviour and contribute to sustainable development.
PEO 4	Effective Communication and Teamwork Demonstrate strong communication skills and teamwork.
PEO 5	Research and Innovation Pursue research and entrepreneurial ventures in computing technologies

2. Program Outcomes (POs) and Program Specific Outcomes (PSOs):

POs	Graduate Attribute	Description
PO1	Computational Knowledge	Understand and apply mathematical, computing, and domain knowledge for conceptualizing computer-based solutions for real-world problems.
PO2	Problem Analysis	Identify, critically analyze, and formulate complex computing problems using the knowledge of computer science and information technology.
PO3	Design and Development of Solutions	Design and develop computer-based systems and components for solving complex real-world problems using appropriate computing technologies.
PO4	Conduct Investigations of Complex Computing Problems	Design and conduct experiments, analyze and interpret data, and provide valid conclusions.
PO5	Modern Tool Usage	Recognize and use modern computing tools, and techniques necessary for innovative computer-based solutions.
PO6	Project Management	Demonstrate engineering and management principles and practices to manage projects in multidisciplinary environments.
PO7	Innovation and Entrepreneurship	Identify opportunities, use innovative ideas, and entrepreneurship mindset to create value and wealth for the betterment of self and society.

PO8	Communication Skills	Communicate effectively in the workplace, collaborate, and contribute to achieve common goals.
PO9	Individual and Team Work	Work as a member or leader in diverse teams in a multidisciplinary environment.
PO10	Professional Ethics	Exhibit the professional attitude and ethics needed for a successful career in industry and academia.
PO11	Societal & Environmental Concerns	Recognize the social and environmental impacts of computer technology and demonstrate professional practices needed for sustainable development.
PO12	Life-long Learning	Recognize the need for and develop the ability to engage in continuous learning for personal and organizational growth.

BCA ARTIFICIAL INTELLIGENCE - PROGRAM SPECIFIC OUTCOMES (PSOs)	
PS01	Analytical Problem Solving-Applied mathematical foundations and algorithmic principles to identify, analyze, and solve complex Artificial Intelligence problems, fostering critical and analytical thinking skills.
PS02	Develop AI-based solutions for business problems, showcasing the ability to understand business requirements and provide innovative AI-driven strategies for optimization and decision-making.
PS03	Ethical and Responsible AI-Understand and apply ethical standards in AI development, considering societal implications. Demonstrate a commitment to responsible AI practices and contribute ethically to industry and societal needs.
PSO4	AI Security Integration: Integrate security measures into AI systems, ensuring the robustness and protection of AI models and data against adversarial attacks and unauthorized access.

II) Course Details

Course Title	:	Natural Language Processing
Course Code	:	24BCAAI504
Credits	:	3
L-T-P-E	:	3-0-0-0
Notional Learning Hours	:	45
Contact Hours	:	45
Semester	:	5
Course Category	:	Core Paper
Contribution to Major/Minor	:	Major
Course Type	:	
CA:ESE	:	50: 50
CA Pass Percentage	:	0
ESE Pass Percentage	:	40%
Aggregate Pass Marks	:	40

1. Nature and Orientation of Course

a. *Percentage of the course that is oriented towards:*

Skills	:	60
Entrepreneurship	:	10
Employability	:	30

b. *Percentage of the course that is relevant towards the development needs*

Globally	:	25
Nationally	:	25
Regionally	:	25
Locally	:	25

c. *Percentage of the course that is oriented towards*

Indian Knowledge Systems	:	20 %
Sustainable Development Goals (SDGs) (Please specify which SDG)	:	
Emerging Technologies (E.g. AI)	:	AI with NLP

2. Course Objectives

COB1	To know the various Natural Language Processing (NLP) techniques.
COB2	Learn various techniques under Natural Language Processing (NLP) to solve language processing problems.
COB3	Develop an appreciation for structures in natural language which computers are confronted with when processing natural language
COB4	To apply the NLP techniques on to various applications

3. Course Outcomes and Bloom's Taxonomy

CO	Course Outcome	BT Level	Attainment Threshold
CO01	Understand the fundamentals of Natural Language Processing (NLP) Models	L2 (Understand)	60
CO02	Analyse and represent text data using various techniques and algorithms	L4 (Analyze)	60
CO03	Implement text classification and information extraction techniques	L3 (Apply)	60
CO04	Evaluate NLP techniques for social-media data to detect sentiment, memes, and misinformation.	L5 (Evaluate)	60
CO05	Design and develop chatbots and e-commerce NLP applications integrating dialog and sentiment modules.	L6 (Create)	60

4. CO: PO Mapping

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO10	PO11	PO12	PS01	PS02	PS03	PS04
C01	3	3	2	-	-	-	-	-	-	-	2	-	3		3	3
C02	3	3	3	2	2	-	-	-	2	-		-	-	-	2	2
C03	3	-	3	-	2	2	-	3	-	-	2	2	3	2	-	-
C04	3	2	3	3	3	3	2	2	3	3	2	2	3	3	3	-
C05	3	2	3	2	3	3	2	3	3	3	2	3	3	-	3	3
Articulation	3	2.5	2.8	2.5	2.5	2.6	2	2.6	2.6	3	2	2.3	3	2.3	2.7	2.6

5. Syllabus

Module	Topics	Contact Hours
I	Introduction to NLP: NLP in real world, language basics, approaches; Building NLP models – Eight steps; Web scraping.	9
II	Text Representation: Vectorization, One-Hot, Bag of Words/N-Grams, TF-IDF, Word Embeddings, Doc2Vec; Text Classification workflow; Naïve Bayes.	9
III	Information Extraction: Applications, Pipeline, Key-phrase extraction, Named Entity Recognition, Entity Disambiguation and Linking.	9
IV	NLP for Social Media: Applications, Challenges, Processing social data, Twitter Sentiment, Meme & Fake News detection.	9
V	Chatbots: Applications, Taxonomy, Dialog Systems; NLP for E-Commerce: Catalog search, Review & Sentiment analysis.	9

Text Books:

1. Daniel Jurafsky & James H. Martin, 'Speech and Language Processing', 3rd Ed., 2023.
2. Delip Rao & Brian McMahan, 'Natural Language Processing with PyTorch', O'Reilly, 2019.

Reference Books:

1. Jacob Eisenstein, 'Introduction to Natural Language Processing', MIT Press, 2019.
2. Christopher D. Manning & Hinrich Schütze, 'Foundations of Statistical NLP', MIT Press, 1999.

E-Resources:

- <https://www.fast.ai/posts/2019-07-08-fastai-nlp.html>
- <https://github.com/oxford-cs-deepnlp-2017/lectures>
- <http://demo.clab.cs.cmu.edu/NLP/>

III) Session Plan

Cohort/Section	BCA – 5th Semester
Faculty Member	Dr Prabhu A

Innovative Pedagogies Planned – if Any
Surprise Quiz, Flipped Classroom, Video presentation, etc..

S e s s i o n n o.	M o d u l e N o .	Topic	L - T - P - E C o m p o n e n t	M o d e o f D e l i v e r y	Pedagog ical Tool / Engage ment	Re adi ng Ma teri al	CO
0	I	Briefing session – TLEP, Course content.		Disc ussi on			
1		Overview of NLP & real-world applications	L	Vide os + Lect ure	Video + Discussion	Juraf sky & Marti n Ch.1- 2	1
2		Basics of language & linguistics for NLP	L	Lect ure	Lecture & Think-Pair- Share	Juraf sky & Marti n Ch.1-	1
3		Approaches to NLP: Rule-based, Statistical, Neural	L	Lect ure	Concept Mapping Activity	Juraf sky & Marti n Ch.1-	1

4		NLP 8-Step Process: Data acquisition	L	Lecture	Guided Discussion	Rao & McMahan Ch.1	1
5		NLP 8-Step Process: Cleaning & preprocessing	L	Lecture with coding demo	Hands-on Demo	Rao & McMahan Ch.1	1
6		NLP 8-Step Process: Modeling & evaluation	L	Lecture	Problem-solving Workshop	Rao & McMahan Ch.1	1
7		Web scraping fundamentals (HTTP, HTML parsing)	L	Lecture with coding demo	Coding Demo	Jurafsky & Martin Ch.1-	1
8		Scraping text with Beautiful Soup & Selenium	L	Discussion and Demonstration	Hands-on Lab	Online Docs	1
9		QUIZ and Activity -1 Briefing	L	Discussion	Discussion	LMS	1
10	II	Vectorization & One-Hot Encoding	L	Lecture	Lecture + Quick Quiz	J&M Ch.6-7	2
11		Bag of Words & N-grams	L	Lecture	Worked Examples	J&M Ch.6-7	2

12		TF-IDF weighting	L	Lecture	Hands-on Coding	J&M Ch.6-7	2
13		Intro to Word Embeddings (Word2Vec, GloVe)	L	Lecture	Video + Discussion	Rao & McMahan Ch.4-5	2
14		Training custom embeddings	L	Lecture	Notebook Lab	Rao & McMahan Ch.4-5	2
15		Doc2Vec representations	L	Lecture	Guided Coding	Rao & McMahan Ch.4-5	2
16		Text classification pipeline overview	L	Lecture	Case Walk-through	Rao & McMahan Ch.4-5	2
17		Naïve Bayes implementation with scikit-learn	L	Lecture	Hands-on Lab	Rao & McMahan Ch.4-5	2
18		LinkedIn Certification Assessment briefing	L	Lecture	Assessment Prep	LinkedIn Learning	2
19	III	Information Extraction (IE) overview & apps	L	Lecture	Lecture + Examples	Eisenstein Ch.11	3
20		IE pipeline & key phrase extraction	L	Discussion and Demo	Interactive Demo	Eisenstein Ch.11	3

21		Named Entity Recognition (NER) basics	L	Lecture	Lecture + Quiz	Eisenstein Ch.11	3
22		NER with spaCy	L	Discussion and Demo	Hands-on Lab	spaCy Docs	3
23		Entity disambiguation techniques	L	Lecture	Group Activity	Eisenstein Ch.11	3
24		Entity linking to knowledge bases	L	Lecture	Coding Lab	Eisenstein Ch.11	3
25		Evaluation metrics for IE tasks	L	Lecture	Problem-solving	Eisenstein Ch.11	3
26		Case study presentations on IE	L	Seminar	Student Presentations	Research Papers	3
27		Assignment discussion & feedback	L	Discussion	Feedback Session	Course LMS	3
28	IV	Social media data & challenges	L	Lecture	Lecture + Brainstorm	J&M Ch.24 + Recent Papers	4
29		Collecting data via Twitter API	L	Lab	Live Coding	Twitter Dev Docs	4

30		Preprocessing noisy social text	L	Lab	Hands-on Cleaning	J&M Ch.24 + Recent Papers	4
31		Fundamentals of sentiment analysis	L	Lecture	Concept Lecture	J&M Ch.24 + Recent Papers	4
32		Transformer-based sentiment (BERT)	L	Lab	Notebook Lab	HuggingFace Docs	4
33		Meme detection basics	L	Lecture	Case Discussion	Research Paper	4
34		Fake news detection techniques	L	Lecture	Group Activity	Research Paper	4
35		Case study analysis on social media NLP	L	Seminar	Team Presentation	Case Dataset	4
36		Reflection & wrap-up discussion	L	Discussion	Panel Discussion	Course Notes	4
37	V	Chatbot applications & taxonomy	L	Lecture	Lecture + Videos	Rao & McMahan Ch.9 + Blogs	5

38		Dialog system architecture	L	Lecture	Interactive Diagram	Rao & McMahan Ch.9 + Blogs	5
39		Intent recognition & slot filling	L	Demonstration	Hands-on Lab	Rao & McMahan Ch.9 + Blogs	5
40		Building dialog flows with Rasa	L	Demonstration	Project Lab	Rasa Docs	5
41		Evaluating chatbot performance	L	Demonstration	Workshop	Rao & McMahan Ch.9 + Blogs	5
42		E-commerce catalog search basics	L	Lecture	Lecture + Demo	Research Paper	5
43		Semantic search & recommendations	L	Demonstration	Coding Lab	Recent Blogs	5
44		Review mining & sentiment in e-commerce	L	Lab	Hands-on Lab	Research Paper	5
45		Capstone case study wrap-up	L	Seminar	Group Presentations	Industry Report	5

IV) Evaluation Plan

Innovative Assessments Planned (If any)

They have to develop a small real time project with theoretical knowledge.

Component	Assessment Type	Formative / Summative	Weightage	CO 1	CO 2	CO 3	CO 4	CO 5	Total
CA1	LinkedIn Certification -1	F	10	5	5	-	-	-	10
CA2	Mid – Term Test -1	F	10	4	4	2	-	-	10
CA3	Preparatory exam	F	10			2	4	4	10
CA4	Min Project and Presentation	F	15	1	1	3	5	5	15
CA5	Attendance	S	5	1	1	1	1	1	5
Total CA			50	11	11	8	10	10	45
ESE		S	50						

CA: Instructions and Rubrics

CA1:

- Student have to complete the course within the stipulated time
- Upon completion the course, they have to compete the exams

- Those are successfully completing the course; they have to submit the report in the drive.

Rubrics:

Criteria	85–100% (Excellent)	65–85% (Very Good)	50–65% (Satisfactory)	40–50% (Limited)	<40% (Poor)
Report Preparation	5	4	3	2	1
Submission on time	5	4	3	2	1
	10	8	6	4	2

CA3: Mini Project

- It's a group of 5 students need to take up a real time project
- The Project concept to be relate with NLP
- After the completion of project, they have to explain the project and present Infront of the students as per the slot.
- During the presentation, all team members need to present.

Rubrics:

Criteria	Quality of Project Work (10)	Team work(2)	Presentation(3))

Course Delivery Details

I) Sessions and Evaluation Details

Semester Start Date	:	20.07.2026
Last Instruction Date	:	
No. of Contact Hours	:	45
Average Attendance	:	75
Date of 1 st CA Component	:	
Date of 2 nd CA Component	:	
Date of 3 rd CA Component	:	
Date of 4 th CA Component	:	

II) Student Engagement and Communication

- Communication with students for the TLEP
- Assessment instructions given to students
- Communication with students for study material/ pre-reads / pre-views/ discussions etc.
- Communication with students of CA marks
- Study material shared
- CA submissions by students (One from the best score and another from an average score). (The sample should capture how the rubrics has been applied during the evaluation.)
- Proof of MOOCs – Swayam / LinkedIn learning etc. (Student certificates)
- ESE Question Paper

II) Student-wise attendance report

III) Remedial Measures (if any)

Rationale for identifying students for Remedial sessions:

List of Students Identified for Remedial

USN	Student Name	Reason for Requiring Remedial

Proposed Remedial Measures:

Proof of Remedial Measures

- Remedial classes timetable
- Remedial Class schedule
- Remedial Assessment details

Course Results and Outcome Attainment

II) Results

[illegible]

III) Result Analysis

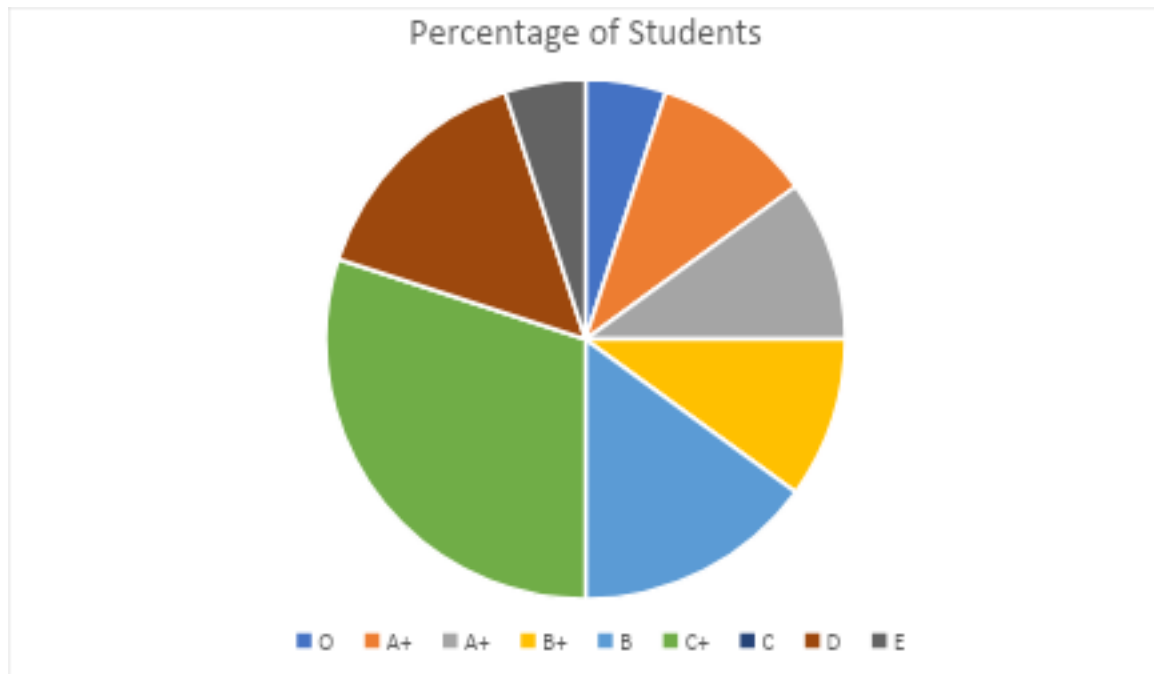
Total Students Enrolled in the Course:

No. of Students appeared for ESE:

No. of Students passed:

Pass Percentage:

Grade		Percentage of Students
O Grade	:	
A+	:	
A	:	
B+	:	
B	:	
C+	:	
C	:	
D	:	
Fail	:	



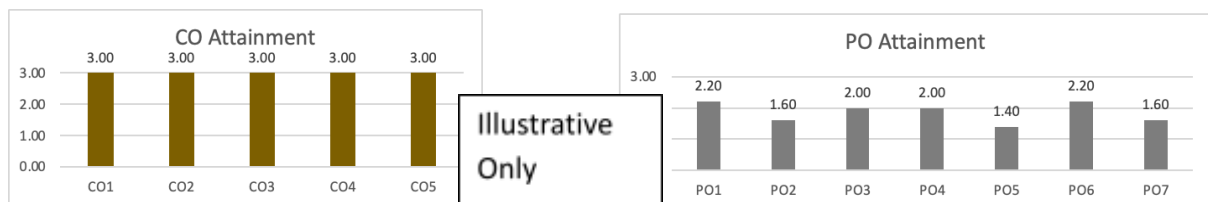
IV) Attainment Analysis

Course	
Course Code	
Faculty	
Section / Cohort	
Section / Cohort Strength	
Semester	

	Level
>=80% percent or more of students score above Threshold*	3
Between 70%-80% percent of students score above Threshold	2
Between 60%-70% percent of students score above Threshold	1
<=60% of students score above the Threshold	0

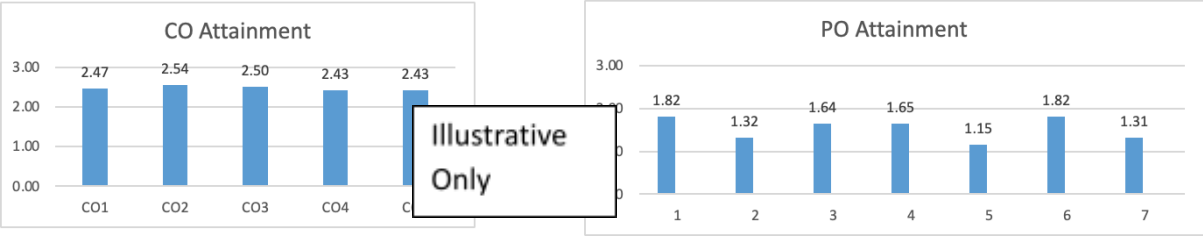
*CO-wise threshold set in the TLEP

Direct Attainment

[illegible]

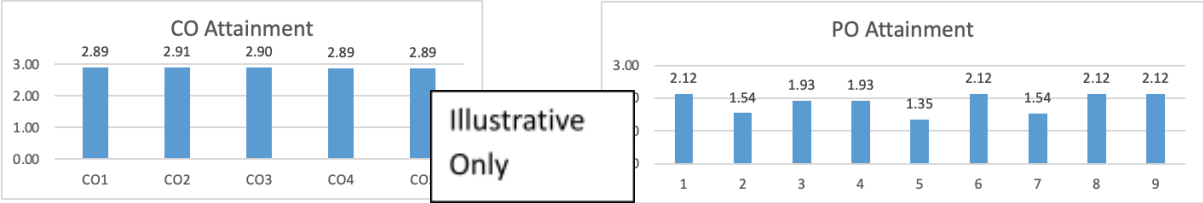
Indirect Attainment

[illegible]



Overall Attainment

Course Outcome	Overall Attainment	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9
C01										
C02										
C03										
C04										
C05										



Course Exit Report and Plan for Improvement

Reflection on the Teaching-Learning Experience
Analysis of Outcome Attainment
CO-wise improvement plan